Artificial Intelligence Technology and Application

Machine Learning Lab Guide

Student Version

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1 Linear Regression

1.1 Introduction

1.1.1 About This Lab

This experiment uses the basic Python code and the simplest data to reproduce how a linear regression algorithm iterates and fits the existing data distribution.

The NumPy and Matplotlib modules are used in the experiment. NumPy is used for calculation, and Matplotlib is used for drawing.

1.1.2 Objectives

Upon completion of this task, you will be able to:

- Be familiar with basic Python statements.
- Master the procedure for implementing linear regression.

1.2 Procedure

1.2.1 Preparing Data

Randomly set ten pieces of data, with the data in a linear relationship.

Convert the data into an array format so that the data can be directly calculated when multiplication and addition are used.

Code:

```
# Import the required modules NumPy for calculation and Matplotlib for drawing.
import numpy as np
import matplotlib.pyplot as plt
#This code is used only for Jupyter Notebook.
%matplotlib inline

# Define data and convert the list into an array.



# Display the data through the scatter chart.
```

Output:

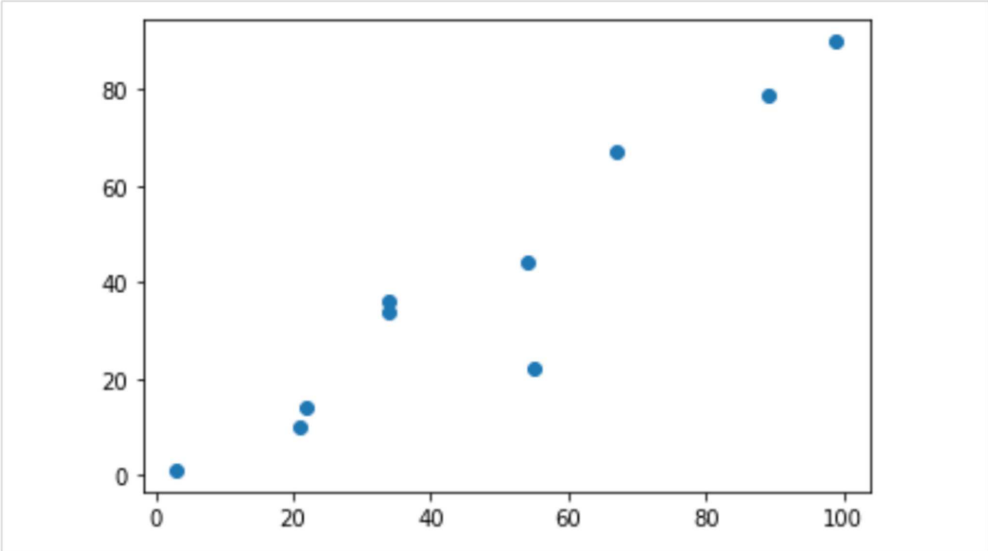

Figure 1-1 Scatter chart

1.2.2 Defining Related Functions

Model function: defines the linear regression model wx+b.

Loss function: calculates the mean square error.

Optimization function: calculates the partial derivatives of w and b by using the gradient descent method.

Code:

```
# The basic linear regression model is wx+b. In this example, the model is ax+b as a two-dimensional space is used.

# The mean square error loss function is the most commonly used loss function in the linear regression model.

# The optimization function mainly uses the partial derivatives to update a and b.

# Perform function iteration to return a and b.
```

1.2.3 Starting Iteration

Step 1 Initialize the iterative optimization model.

Code:

```
# Initialize and display parameters.
```

Output:

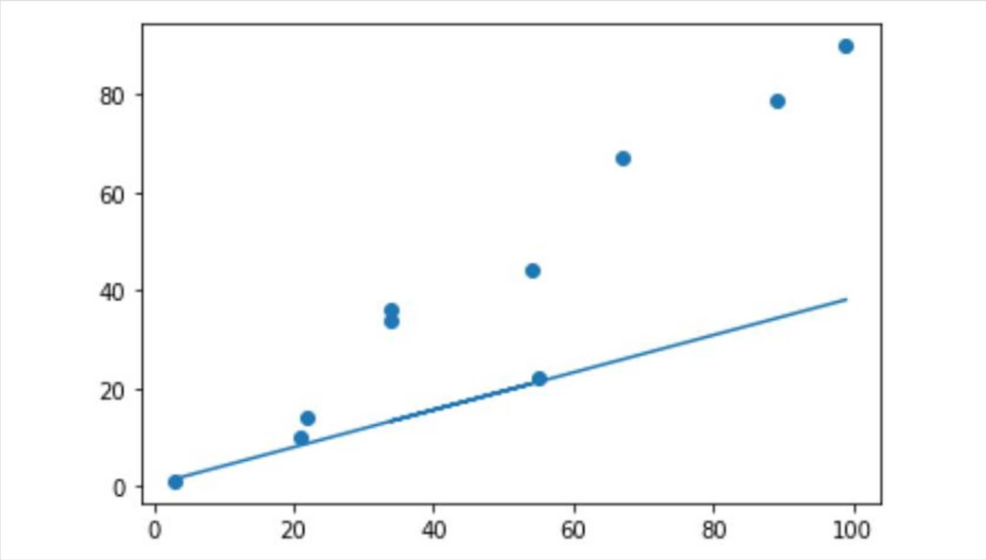

Figure 1-2 First iteration

Step 2 Perform the second iteration and display the parameter values, loss values, and visualization effect.

Code:

```

```

Output:

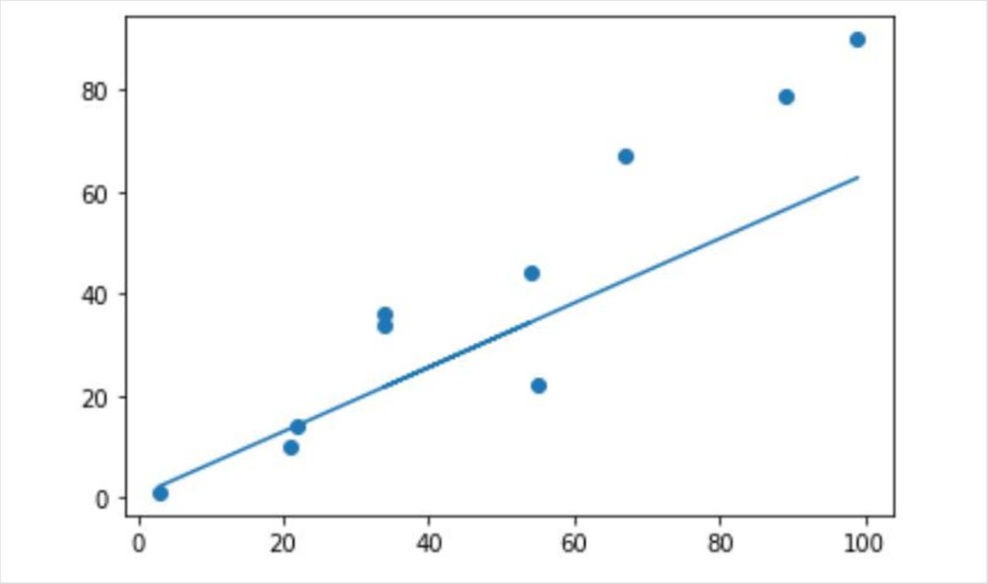

Figure 1-3 Second iteration

Step 3 Perform the third iteration and display the parameter values, loss values, and visualization effect.

Code:

Output:

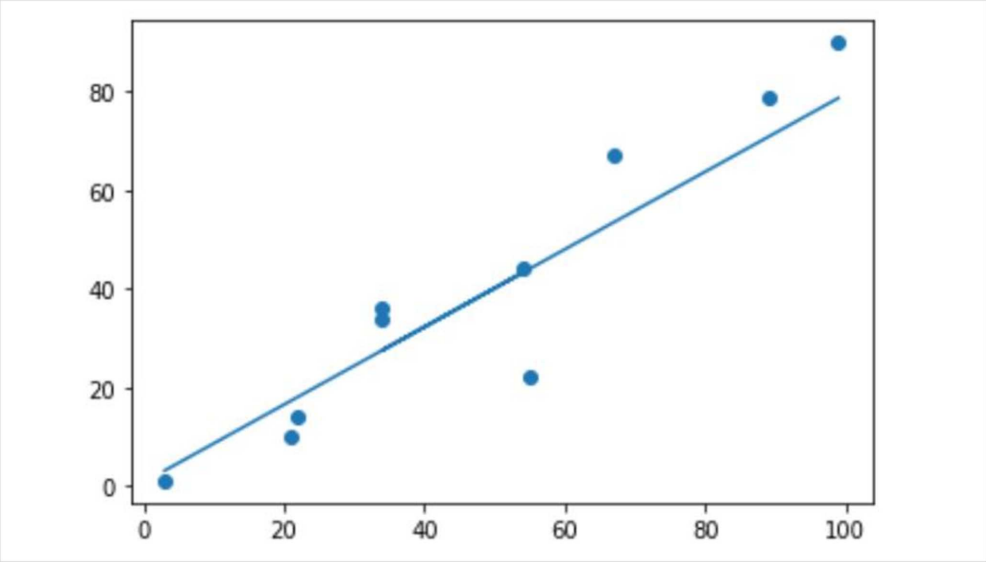

Figure 1-4 Third iteration

Step 4 Perform the fourth iteration and display the parameter values, loss values, and visualization effect.

Code:

Output:

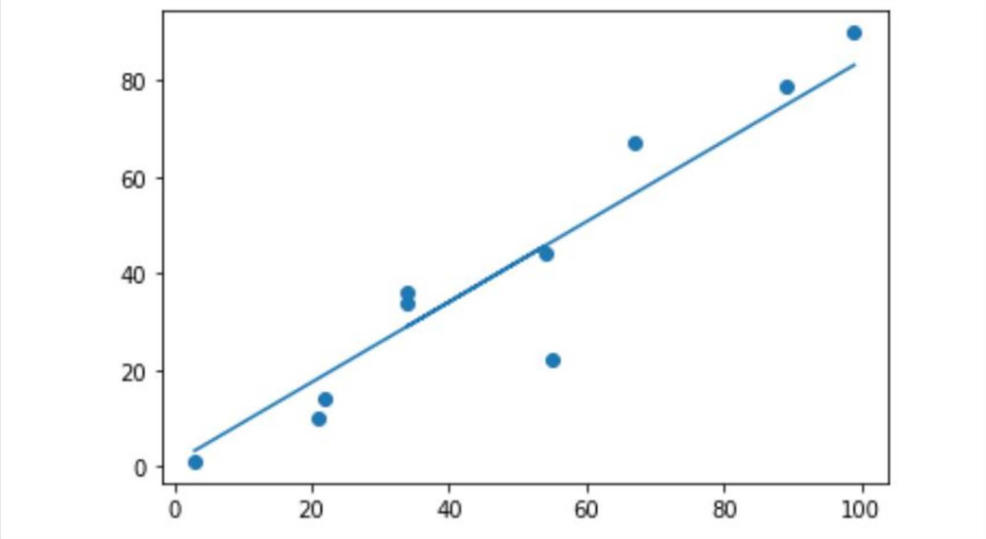

Figure 1-5 Fourth iteration

Step 5 Perform the fifth iteration and display the parameter values, loss values, and visualization effect.

Code:

Output:

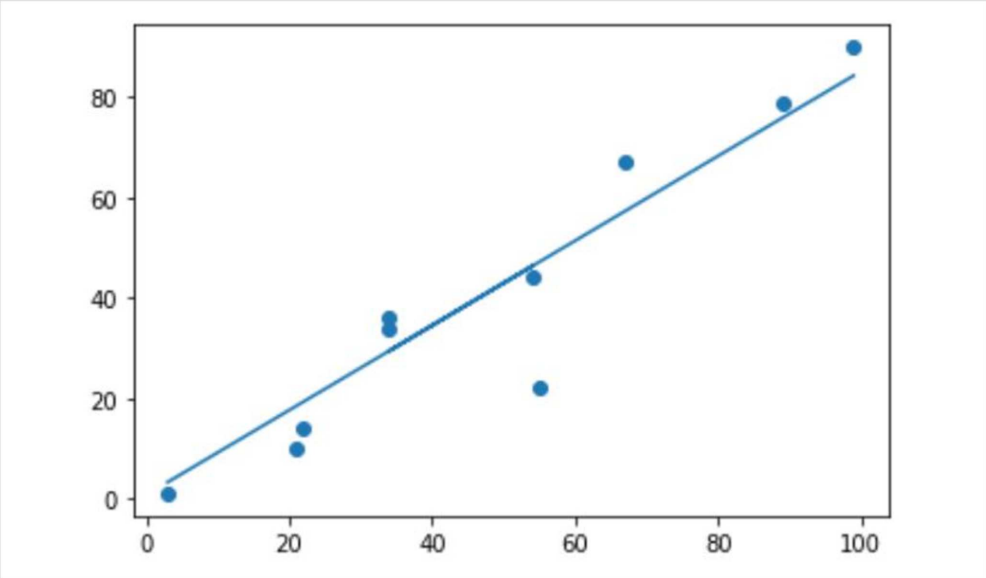

Figure 1-6 Fifth iteration

Step 6 Perform the 10000th iteration and display the parameter values, loss values, and visualization effect.

Code:

Output:

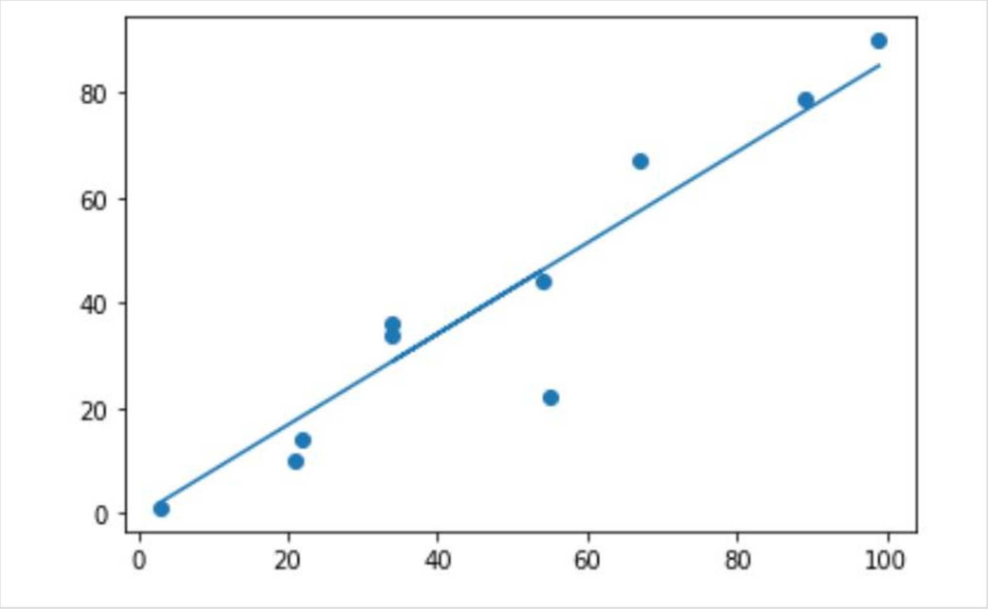

Figure 1-7 10000th iteration

----End

1.3 Thinking and Practices

1.3.1 Question 1

Must the loss value return to zero when the raw data is modified?

1.3.2 Question 2

What is the function of **Lr** during **Lr** modification?